

8. Develop a swing application using JList, JTree, JTable, JTabbedPane and JScrollPane.

Type	Input	Expected Output	Explanation
Success	Select item in JList	Selected item displayed	JList validation
Success	Expand node in JTree	Child nodes displayed	JTree validation
Success	Show records in JTable with JScrollPane	Table displayed with scroll	JTable and scrollpane
Failure/Edge	Empty list/table	Component shown without crash	Empty data
Failure/Edge	Large table data	ScrollPane works correctly	UI scalability

Algorithm:

1. **Start** the program.
2. Create a JFrame and set its title, size, layout, and close operation.
3. Create a JTabbedPane with three tabs: **JList, JTree, JTable**.
4. **JList**: Create a list of fruits and display the selected item using a ListSelectionListener.
5. **JTree**: Create a tree with Courses as the root and Java and Python as child nodes.
6. **JTable**: Create a table with student ID, name, and department.
7. Add all three panels to the tabbed pane.
8. Add the tabbed pane to the JFrame and make it visible.
9. **Stop** the program when the window is closed.

Program:

```
import javax.swing.*.*;
import javax.swing.event.*;
import javax.swing.tree.DefaultMutableTreeNode;
import java.awt.*.*;

public class SwingComponentsDemo extends JFrame {

    private JLabel listLabel;

    public SwingComponentsDemo() {
        setTitle("Swing Components Demo");
        setSize(700, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JTabbedPane tabbedPane = new JTabbedPane();
```

```
// ===== JLIST TAB =====
JPanel listPanel = new JPanel(new BorderLayout());

String[] fruits = {"Apple", "Banana", "Mango", "Orange", "Grapes"};
JList<String> jList = new JList<>(fruits);

listLabel = new JLabel("Select an item from the list");
jList.addListSelectionListener(new ListSelectionListener() {
    @Override
    public void valueChanged(ListSelectionEvent e) {
        if (!e.getValueIsAdjusting()) {
            listLabel.setText("Selected Item: " + jList.getSelectedValue());
        }
    }
});

listPanel.add(new JScrollPane(jList), BorderLayout.CENTER);
listPanel.add(listLabel, BorderLayout.SOUTH);

// ===== JTREE TAB =====
JPanel treePanel = new JPanel(new BorderLayout());

DefaultMutableTreeNode root =
    new DefaultMutableTreeNode("Courses");

DefaultMutableTreeNode javaNode =
    new DefaultMutableTreeNode("Java");
javaNode.add(new DefaultMutableTreeNode("Swing"));
javaNode.add(new DefaultMutableTreeNode("JDBC"));

DefaultMutableTreeNode pythonNode =
    new DefaultMutableTreeNode("Python");
pythonNode.add(new DefaultMutableTreeNode("Django"));
pythonNode.add(new DefaultMutableTreeNode("Flask"));

root.add(javaNode);
root.add(pythonNode);

JTree tree = new JTree(root);

treePanel.add(new JScrollPane(tree), BorderLayout.CENTER);

// ===== JTABLE TAB =====
JPanel tablePanel = new JPanel(new BorderLayout());

String[] columns = {"ID", "Name", "Department"};
```

```

Object[][] data = {
    {"101", "John", "CSE"},
    {"102", "Alice", "ECE"},
    {"103", "David", "IT"},
    {"104", "Sophia", "ME"},
    {"105", "James", "EEE"},
    {"106", "Emma", "CSE"},
    {"107", "Robert", "IT"},
    {"108", "Olivia", "ECE"}
};

JTable table = new JTable(data, columns);

JScrollPane tableScrollPane = new JScrollPane(table);

tablePanel.add(tableScrollPane, BorderLayout.CENTER);

// ===== ADD TABS =====
tabbedPane.addTab("JList", listPanel);
tabbedPane.addTab("JTree", treePanel);
tabbedPane.addTab("JTable", tablePanel);

add(tabbedPane);

setVisible(true);
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new SwingComponentsDemo());
}
}

```